Algebra 2 Review: Quadratic Functions & Complex Numbers

Answer Key

Station 1: Quadratic Functions, Forms, and Graphs

1. $f(x) = 2(x-2)^2 - 5$; vertex $(2, -5)$; axis $x = 2$; minimum value -5
   $2(x^2 - 4x + 4) - 8 + 3$
2. $y = -(x+3)(x-1) = -x^2 - 2x + 3$; vertex $(-1, 4)$
   $a(0+3)(0-1) = 3 \Rightarrow a = -1$; *axis is* $x = \dfrac{-3+1}{2}$
3. reflect over the x-axis, vertical compression by $\frac{1}{2}$, left 3, down 2; vertex $(-3, -2)$; opens down
   $x + 3$ *shifts left, not right*
4. $f(x) = 2(x-5)(x+3)$; intercepts 5 and -3; axis $x = 1$; vertex $(1, -32)$
   axis is halfway between the intercepts
5. $y = 2(x+2)(x-5) = 2x^2 - 6x - 20$
   $-24 = a(3)(-4) \Rightarrow a = 2$
6. $y = -2(x+3)^2 + 2$
   $-6 = a(2)^2 + 2 \Rightarrow a = -2$
7. f: minimum 5 at $x = 5$; g: minimum 1 at $x = 2$; f has the greater minimum
   the table is symmetric about $x = 2$
8. (a) $y \le 11$; (b) $-7 \le y \le 9$ (still increasing on $[0, 2]$, so use the endpoints)
   vertex is $(3, 11)$, *outside* $[0, 2]$
9. $y = 2x^2 - 3x + 1$
   $c = 1$; *then* $a - b = 5$ *and* $2a + b = 1$
10. $y = \dfrac{1}{2}(x-2)^2 - 3 = \dfrac{1}{2}x^2 - 2x - 1$
    $-1 = a(0-2)^2 - 3 \Rightarrow a = \dfrac{1}{2}$

Station 2: Solving Quadratic Equations and the Discriminant

1. $x = \dfrac{4}{3}$ or $x = -\dfrac{1}{2}$
   $(3x - 4)(2x + 1) = 0$
2. $x = -1 \pm 3\sqrt{2}$
   $(x+1)^2 = 18$ *and* $\sqrt{18} = 3\sqrt{2}$
3. $x = -5 \pm 2\sqrt{7}$
   $x^2 + 10x + 25 = 3 + 25$; $(x+5)^2 = 28$
4. $x = 3 \pm \dfrac{\sqrt{22}}{2}$
   $(x-3)^2 = \dfrac{11}{2}$
5. $x = \dfrac{-3 \pm \sqrt{65}}{4}$
   $b^2 - 4ac = 9 + 56 = 65$
6. (a) 0: one repeated real solution; (b) -39: two nonreal complex solutions; (c) 49: two rational real solutions
   in (c), $b = -1$ *so* $b^2 = 1$
7. $b^2 - 4ac = 16 - 4k$: (a) $k < 4$; (b) $k = 4$; (c) $k > 4$
8. $x = \pm 2$ or $x = \pm 3$
   $(u-4)(u-9) = 0$
9. $x = 5$ or $x = -4$
   move everything to one side first: $x^2 - x - 20 = 0$
10. $x = 3$ (since $x = -2$ is extraneous: $\sqrt{4} = 2 \neq -2$)
    squaring gives $x^2 - x - 6 = 0$

Station 3: Complex Numbers

1. (a) $4i\sqrt{3}$; (b) $(3i)(2i) = 6i^2 = -6$
   write each root with i *before multiplying*
2. (a) i; (b) -1; (c) i
   divide the exponent by 4 and use the remainder; $\dfrac{1}{i^3} = \dfrac{1}{-i} = i$
3. $1 + 6i$
   distribute the minus sign to both parts of $2 - 8i$
4. $26 + 7i$
   $6 - 8i + 15i - 20i^2$
5. $7 - 24i$
   $16 - 24i + 9i^2$; *do not square each term only*
6. conjugate $-2 + 7i$; product 53
   $(-2)^2 + 7^2$
7. $-\dfrac{2}{13} + \dfrac{29}{13}i$
   multiply by $\dfrac{3+2i}{3+2i}$; *denominator* $9 + 4 = 13$
8. $x = 2 \pm i\sqrt{5}$
   $\dfrac{4 \pm \sqrt{-20}}{2}$
9. $x = \dfrac{-1 \pm i\sqrt{14}}{3}$
   $\dfrac{-2 \pm \sqrt{-56}}{6}$ *and* $\sqrt{-56} = 2i\sqrt{14}$
10. other root $3 + 4i$; $x^2 - 6x + 25 = 0$
    sum 6, *product* $9 + 16 = 25$

Station 4: Quadratic Models, Inequalities, and Systems

1. maximum 144 ft at $t = 2$ s; lands at $t = 5$ s (reject $t = -1$)
   $-16(t-5)(t+1) = 0$
2. $R(p) = 500p - 20p^2$; price \$12.50; maximum revenue \$3,125
   vertex at $p = \dfrac{-500}{2(-20)}$
3. 40 ft by 60 ft; maximum area 2,400 ft^2
   $3x + 2L = 240$, *so* $A = x \cdot \dfrac{240 - 3x}{2}$
4. $3 < x < 11$: between 300 and 1,100 items
   $P = -2(x-3)(x-11)$
5. (a) $(-\infty, -2] \cup [5, \infty)$; (b) $(-1, 5)$
   in (b), multiply by -1 *and reverse the inequality*
6. $y < -x^2 + 4$; $(0,0)$: $0 < 4$ is true, so yes; $(3,0)$: $0 < -5$ is false, so no
   dashed boundary means strict inequality
7. $(1, -2)$ and $(5, 6)$
   $x^2 - 6x + 5 = 0$, *so* $(x-1)(x-5) = 0$
8. (a) one: $(3, 2)$; (b) none; (c) two: $(1, 2)$ and $(5, 10)$
   discriminants of $x^2 - 6x + c$: 0, -24, 16
9. $y = -1.9x^2 + 10.4x + 4.8$; maximum ≈ 19.0 ft at $x \approx 2.7$ ft
   $x = \dfrac{-10.4}{2(-1.9)}$
10. $y = -\dfrac{3}{40}x^2 + 30$; height 22.5 ft; width $20\sqrt{2} \approx 28.3$ ft
    $0 = a(400) + 30 \Rightarrow a = -\dfrac{3}{40}$